Free balancing equations worksheet worksheet 6

Maths | 9th grade | Algebra

1. If 3x - 7 = 2(x + 4), what is the value of x?

A) -1
B) 5
C) 15
D) 8

2. Simplify the expression: 4(2y - 5) + 3(3y + 2)

A) 17y - 14
B) 14y - 14
C) 7y + 1
D) 3y - 14

3. Solve for z: 5 - 2(z + 3) = 3z - 1

A) 2
B) -2
C) 1
D) -1

4. The sum of three consecutive integers is 51. Find the middle integer.

A) 16
B) 17
C) 18
D) 19

5. Evaluate the expression when a = -3 and b = 2: 2a^2 - 3b + 4

A) 12
B) 10
C) 14
D) 6

6. If f(x) = 2x^2 - 3x + 5, what is f(-1)?

A) 10
B) 6
C) 0
D) 4

7. What is the solution set for the inequality 3(x - 2) > 2x + 5?

A) x < 11
B) x > -11
C) x > 11
D) x < -11

8. Find the value of k if 2x + 3y = 8 and 4x + 6y = k are equivalent equations.

A) 16
B) 8

C) 4
D) 0

9. If x is a number such that x/2 + 5 = 3x - 1, what is the value of x?

A) 3
B) 2
C) 4
D) 5

10. Simplify the expression: (3x^2 - 2x + 4) - (x^2 + 4x - 1)

A) 2x^2 - 6x + 5
B) 4x^2 - 6x + 3
C) 2x^2 - 6x + 3
D) 2x^2 + 6x + 3

Answer Key

1. B
Distribute on the right: 3x - 7 = 2x + 8. Subtract 2x from both sides: x - 7 = 8. Add 7 to both sides: x = 15.

2. A
Distribute and combine like terms: 8y - 20 + 9y + 6 = 17y - 14.

3. C
Distribute and solve: 5 - 2z - 6 = 3z - 1. Simplify: -2z - 1 = 3z - 1. Add 2z to both sides: -1 = 5z - 1. Solve for z: z = 1.

4. B
Let the integers be x, x+1, x+2. Their sum is 3x + 3 = 51. Simplify: 3x = 48. Solve: x = 16. The middle integer is x + 1 = 17.

5. C
Substitute: 2(-3)^2 - 3(2) + 4 = 2*9 - 6 + 4 = 18 - 6 + 4 = 16.

6. A
Substitute: f(-1) = 2(-1)^2 - 3(-1) + 5 = 2*1 + 3 + 5 = 10.

7. C
Distribute: 3x - 6 > 2x + 5. Subtract 2x: x - 6 > 5. Add 6: x > 11.

8. A
Multiply the first equation by 2: 2(2x + 3y) = 2(8), resulting in 4x + 6y = 16. Thus, k = 16.

9. C
Isolate x: x/2 + 5 = 3x - 1, multiply through by 2: x + 10 = 6x - 2, solve: 12 = 5x, x = 4.

10. C
Subtract: 3x^2 - 2x + 4 - x^2 - 4x + 1, result: 2x^2 - 6x + 5.